# Proximal NND Fitting with Known Heteroskedastic Errors and Unknown Homoskedastic Noise

## 1 Model

Let $\Omega \subseteq \{1, \dots, m\} \times \{1, \dots, n\}$ denote the observed entries. Suppose each observed entry has a known heteroskedastic standard error $s_{ij}$, but that these known errors may miss an additional homoskedastic variance component $g = \gamma^2$. The likelihood is

$$Y_{ij} \mid X_{ij}, g \sim \mathcal{N}\left(X_{ij},\, s_{ij}^2 + g\right), \qquad (i,j) \in \Omega. \tag{1}$$

Missing entries are handled by omitting them from the likelihood.

We use a nuclear-norm distribution prior on $X$. For fitting, it is most stable to parameterize the grid directly in terms of the effective nuclear-norm rate

$$\eta = \lambda_{\text{eff}},$$

rather than a base rate $\lambda$ coupled to $g$. Thus

$$X \mid \eta \sim \text{NND}(\eta), \qquad p(X \mid \eta) \propto \eta^{mn} \exp\{-\eta \|X\|_*\}. \tag{2}$$

If a base homoskedastic parameterization is desired for reporting, it can be recovered after fitting as

$$\lambda = \eta \sqrt{g}. \tag{3}$$

Under the effective-rate parameterization, the NND normalizer contributes $-mn \log \eta$ to the negative log posterior but does not add an $(mn/2) \log g$ term. This substantially reduces the feedback between the regularization path and the variance update.

For $g$ we use an optional inverse-gamma prior,

$$p(g) \propto g^{-(a_0+1)} \exp(-b_0/g), \tag{4}$$

but the fitting strategy below only requires a one-dimensional prior penalty in $g$.

## 2 Taylor Approximation for the Variance Update

Let $\mu$ denote the current matrix estimate. Define the row-wise effective variances

$$v_{ij}(g) = s_{ij}^2 + g, \qquad p_{ij}(g) = \begin{cases} 1/v_{ij}(g), & (i,j) \in \Omega, \\ 0, & (i,j) \notin \Omega, \end{cases} \tag{5}$$

and the nuclear-norm curvature proxy

$$B(\mu) = \left(\mu^\top \mu + \epsilon I\right)^{-1/2}. \tag{6}$$

For fixed $(\mu, g, \eta)$, the Taylor row precision is

$$A_i(\mu, g, \eta) = \mathrm{diag}\{p_{ij}(g)\}_{j=1}^n + \eta B(\mu). \tag{7}$$

The Taylor approximation therefore defines a full row covariance

$$\Sigma_i(\mu, g, \eta) = A_i(\mu, g, \eta)^{-1}. \tag{8}$$

This covariance is dense in general because $B(\mu)$ is dense. For entrywise scores and lightweight diagnostics we usually retain only the marginal variances,

$$\mathrm{diag}(\Sigma_i) = \mathrm{diag}\big(A_i(\mu, g, \eta)^{-1}\big). \tag{9}$$

This diagonal extraction is a storage and scoring approximation, not an assumption that the row covariance itself is diagonal.

Holding $\mu$ fixed, the collapsed Taylor objective for updating $g$ is, up to constants independent of $g$,

$$\begin{aligned}\mathcal{J}_g(g \mid \mu, \eta) = \frac{1}{2} \sum_{(i,j)\in\Omega} \left[\log\{s_{ij}^2 + g\} + \frac{(y_{ij} - \mu_{ij})^2}{s_{ij}^2 + g}\right] + \frac{1}{2} \sum_{i=1}^m \log |A_i(\mu, g, \eta)| \\ + \eta \|\mu\|_* + (a_0 + 1) \log g + \frac{b_0}{g}.\end{aligned} \tag{10}$$

The first line is the heteroskedastic Gaussian likelihood plus the Taylor log-determinant correction. The second line is the fixed-rate NND prior and the optional inverse-gamma prior on $g$. The $\eta\|\mu\|_*$ and $-mn \log \eta$ NND-normalizer terms are constant with respect to $g$, but they are retained when scoring a grid of $\eta$ values.

### 2.1 Stochastic Hutchinson–CG Update for $g$

The exact update of $g$ is still expensive because each evaluation of (10) or its derivative requires the term

$$\frac{1}{2} \sum_{i=1}^m \log |A_i(\mu, g, \eta)|. \tag{11}$$

Naively this costs one Cholesky factorization of an $n \times n$ matrix for each row. Since $g$ is scalar, a practical alternative is to update $\ell = \log g$ using a stochastic gradient. The likelihood and variance-prior parts have closed-form derivatives. Only the log-determinant derivative is estimated stochastically.

For a fixed row $i$,

$$\frac{\partial}{\partial \ell} \log |A_i| = \mathrm{tr}\left[A_i^{-1} \frac{\partial A_i}{\partial \ell}\right], \qquad \ell = \log g. \tag{12}$$

Using Hutchinson probes $z_{ir}$ with $\mathbb{E}(z_{ir} z_{ir}^\top) = I$, this trace can be estimated by

$$\mathrm{tr}\left[A_i^{-1} \frac{\partial A_i}{\partial \ell}\right] \approx \frac{1}{R} \sum_{r=1}^R z_{ir}^\top A_i^{-1} \frac{\partial A_i}{\partial \ell} z_{ir}. \tag{13}$$

For the effective-rate parameterization,

$$\frac{\partial p_{ij}}{\partial \ell} = -\frac{g}{(s_{ij}^2 + g)^2}, \qquad (i,j) \in \Omega, \tag{14}$$

$$\frac{\partial}{\partial \ell}\eta = 0. \tag{15}$$

Therefore

$$\frac{\partial A_i}{\partial \ell} = \operatorname{diag}\left\{-\frac{g}{(s_{ij}^2 + g)^2}\mathbf{1}\{(i,j) \in \Omega\}\right\}_{j=1}^{n}. \tag{16}$$

Each Hutchinson term only requires solving a linear system

$$A_i x_{ir} = \frac{\partial A_i}{\partial \ell} z_{ir}. \tag{17}$$

These solves are performed with conjugate gradients using matrix-vector products

$$x \mapsto \operatorname{diag}\{p_{ij}(g)\}_{j=1}^{n} x + \eta B(\mu) x. \tag{18}$$

The row/probe solves are independent and can be vectorized over both $i$ and $r$. In implementation we keep the Hutchinson probes fixed during the fit and warm start each CG solve from the previous solution for the same row/probe. Fixed probes reduce stochastic jitter in the one-dimensional trajectory for $g$, while warm starts exploit the fact that successive $g$ values usually change slowly.

The resulting stochastic update is

$$\ell^{(t+1)} = \min\left\{\ell_{\max}, \max\left[\ell_{\min}, \ell^{(t)} - \rho_t \widehat{\nabla}_\ell \mathcal{J}_g(e^{\ell^{(t)}} \mid \mu, \eta)\right]\right\}, \tag{19}$$

so the log-variance stays in the configured interval $[\ell_{\min}, \ell_{\max}]$. The gradient may also be clipped for stability. This update is no longer a monotone backtracking step because the log-determinant gradient is stochastic, but in benchmarks a small fixed learning rate with a handful of probes closely tracked the exact Cholesky update while substantially reducing wall-clock time.

## 3 Efficient Proximal Fit for the Matrix

The full Taylor objective would include the log-determinant term in the update for $\mu$, which is costly. For efficiency, we instead update $\mu$ using the weighted nuclear-norm proximal problem

$$\hat{\mu}(g, \eta) = \arg\min_X \left\{\frac{1}{2}\sum_{(i,j)\in\Omega} \frac{(y_{ij} - X_{ij})^2}{s_{ij}^2 + g} + \eta \|X\|_*\right\}. \tag{20}$$

This subproblem is solved by FISTA or proximal gradient with singular-value thresholding. The Taylor approximation is then used only to update $g$ and to score candidate $\eta$ values. This is substantially cheaper than including the Taylor log-determinant in every gradient step for $X$.

The alternating fixed-$\eta$ fitting strategy is:

1. Initialize $g > 0$.

2. With $g$ fixed, solve (20) for $\mu$.
3. With $\mu$ fixed, update $\ell = \log g$ by a one-dimensional backtracking step on (10).
4. Repeat until $g$ and $\mu$ stabilize, or until a fixed outer-iteration budget is reached.

The effective penalty returned by the fit is

$$\hat{\lambda}_{\text{eff}} = \eta, \tag{21}$$

which is the quantity directly comparable to a known-variance proximal nuclear-norm penalty.

# 4 Selecting $\eta = \lambda_{\text{eff}}$

In practice, $\eta$ should be selected by a grid or other outer search. For each candidate $\eta_k$, fit $(\hat{\mu}_k, \hat{g}_k)$ using the fixed-$\eta$ alternating scheme above. Candidate fits can then be scored by one of the following criteria.

**Entrywise cross-validation.** Hold out a subset of observed entries and choose the $\eta_k$ with best predictive error or predictive likelihood on the held-out entries.

**Taylor collapsed objective.** Choose the candidate with the smallest final approximate objective, including the fixed-rate NND normalizer and the Taylor log-determinant correction.

**Analytical LOO.** Use the marginal variances $\operatorname{diag}(\hat{\Sigma}_i)$ extracted from the full row covariances and evaluate the Gaussian leave-one-entry predictive density with effective variances $s_{ij}^2 + \hat{g}_k$.

The Taylor and LOO scores are model-internal criteria, while cross-validation is an external predictive criterion. The internal criteria are much cheaper than nested CV when a fitted grid is already available, but their performance should be checked against CV in regimes of interest.

# 5 Warning on Joint Estimation of $g$ and a Base $\lambda$

Joint conditional-mode updates for $(g, \lambda)$ are not recommended for this approximation. Under the scaled prior, the conditional mode for the base parameter $\lambda$ has the schematic form

$$\hat{\lambda} \approx \frac{mn + a_\lambda - 1}{\|\mu\|_*/\sqrt{g} + b_\lambda}. \tag{22}$$

Thus increasing $g$ can increase the base $\lambda$, while a larger base $\lambda$ can in turn affect the next $g$ update through both $\lambda/\sqrt{g}$ and the prior normalizer. Empirically, this feedback can make $\lambda$ and $g$ chase each other and run to boundary values.

For this reason, the recommended workflow is to hold the effective penalty $\eta$ fixed within each fit, estimate $g$ conditional on that $\eta$, and select $\eta$ using a grid scored by CV, Taylor ELBO/objective, or LOO. A base rate $\lambda = \eta\sqrt{\hat{g}}$ can be reported afterward if desired. Joint base-lambda–variance updates should be treated only as a diagnostic or experimental option.